

The Demands Of Successful NEW AUTOMOTIVE PRODUCTS

As vehicles become increasingly software-driven, success will depend on how quickly ideas can move from concept to production. Discover how automotive new product introduction (NPI) is helping manufacturers turn bold ideas into production-ready innovations with greater speed and confidence.

Engineering & Design Bottlenecks



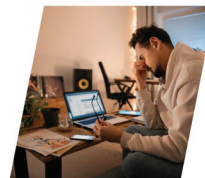
Late design changes causing rework



Lack of digital continuity from design to production



Prototyping cost and time



Inadequate early-stage testing



Legacy systems and siloed data

Solutions

Use of Digital Twins and simulation early in design

Cross-domain engineering collaboration platforms

Virtual validation using CAE tools

Rapid prototyping with additive manufacturing

Predictive analytics for risk identification.

www.tvselectronics.in



VIJAY ADITYA RAGHAVAN is VP, TVS Electronics Limited

My conversations with individuals throughout the automotive ecosystem are consistently linked to a single theme. It is not only the electric vehicle revolution that we are witnessing; we are in the midst of a complete transformation of vehicles, their construction, support, and the overall user experience. This is a change I witness daily as a member of TVS Electronics. The mobility shift is unfolding before our eyes through our work on EV charging infrastructure.

We design charging systems at TVS Electronics, ranging from 300W chargers for two-wheelers to 360kW charging stations that serve large bus fleets. That range tells its own story. This is

no longer a niche market. It is a diverse and fast-growing ecosystem. The pace of change requires us to re-examine how we manage new product introduction (NPI).

The vehicle has become a software platform

Let's start with the product itself and explore where automotive NPI is headed. Today, cars are software-first machines. I still remember learning that the amount of code in a modern high-end passenger car exceeds that found in a fighter aircraft. I found it hard to believe until I experienced it first-hand.

My car is a Suzuki XL6, and the company registered me on its Suzuki Connect platform. Initially, it provided